

ERIK NORRIS

Sr. Staff Mechanical Engineer / Designer | Forensic Architect

erik@eriknorris.com
Redwood City, CA
[resume.eriknorris.com](mailto:erik@eriknorris.com)
eriknorris.com

// AUTODIDACT TECHNICAL POLYGLOT | CONCEPT TO PRODUCTION | BBQS TO DENTAL CHAIRS

PROFESSIONAL SUMMARY

I architect the hardware interface for the physical world—from the heat of a Fissler BBQ to the surgical precision of a KaVo Dental Chair. My career is defined by a relentless drive to translate 'Concept' into 'Mass Production' across every conceivable domain. I have helped validate the most beautiful PC ever made (SGI Indigo), engineered the Gold Standard for pro audio consoles (Avid D-Control), and optimized the OG gaming platform (Xbox). Whether it's managing the 150-watt thermal load of a retinal headset, ensuring IP69K washdown compliance for food robotics, or guiding a Class III cardiac catheter through the FDA—the physics change, but the mission remains the same: Zero Yield Loss. Zero Cosmetic Defects. Total Functional Integrity.

CORE COMPETENCIES

ENGINEERING

- Forensic Engineering
- Mechanical Architecture
- Complex Surface Modeling (Pro/E, Creo)
- Mechanism Design
- Thermal Management (Active/Passive)

MANUFACTURING

- New Product Introduction (NPI)
- Design for Manufacturing (DFM)
- Root Cause Analysis & Yield Improvement
- Injection Molding (Class A)
- Die Casting & Thixomolding
- Sheet Metal (Progressive)

TOOLS & REGULATORY

- PTC Creo / Pro/ENGINEER
- PLM (Windchill Admin)
- Onshape / SolidWorks
- GD&T / Tolerance Analysis
- Technical Program Management
- Supplier Qualification & Strategy
- Regulatory Compliance (UL/FCC/CE)
- Class III Medical Device Standards

PROFESSIONAL EXPERIENCE

HYPHEN | Mechanical Engineer

2021 - 2022

| *Lead engineer for the 'Factory-in-a-Box' food robotics platform.*

- Integrated 350+ electromechanical actuators into a constrained chassis.
- Achieved **IP69K compliance** for wash-down environments, designing custom sealing solutions for doors, cable pass-throughs, and sensor arrays.
- Reduced **Mean Time To Repair (MTTR)** from hours to minutes by architecting a hot-swappable module strategy that decoupled dispensers from the main chassis.

NOON HOME | Senior Mechanical Engineer

2017 - 2018

| Lead mechanical architect for the 'Haptic Switch' smart lighting controller.

- Designed a floating flexure system to isolate the OLED display stack from mounting forces, resulting in **zero cosmetic yield loss** at assembly.
- Collaborated with ID to achieve 'dead-front' aesthetics using precision tint-matching and seamless glass integration.

AVEGANT | Senior Mechanical Engineer

2015 - 2017

| Mechanical Lead for mixed-reality retinal imaging headsets.

- **Thermal Engineering:** Utilized the magnesium headband as a passive heat sink to dissipate 150W equivalent heat load, resolving a critical 'Hot Head' user complaint.
- **Mechanism Design:** Engineered a high-precision IPD adjustment mechanism, ensuring optical alignment stability under drop-test conditions.

KALEIDESCAPE | Senior Mechanical Design Engineer

2008 - 2015

| Sole Mechanical Engineer responsible for the M500, M300, and Cinema One product lines.

- **Crisis Management:** Detected 'Flow Mark' defects in 1,200 parts (Cinema One). Exercised **Line-Down** authority to reject inventory and establish 'Apple-tier' cosmetic standards.
- **Cost Reduction:** Redesigned the 'Macduff' chassis lid, consolidating three parts into one. Reduced assembly time by **15 minutes** and eliminated fastener count by 30%.
- **System Integration:** Integrated slot-loading optical drives (Blu-ray) with custom vibration isolation dampers to prevent skip/read errors.

DIGIDESIGN (AVID) - SC48 | Lead Mechanical Engineer | Project Lux

2007 - 2008

| Engineered a compact, embedded Live Sound console integrating I/O and host processing into a single rack-less shell.

- **The Meltdown Fix:** Eliminated a 100% show-stopping thermal failure (75°C CPU shutdown) by engineering a custom 4U side-intake architecture that stabilized temperature rise to **22.6°C**.
- **Cost Engineering:** Reduced mechanical material costs to **20%** of total COGS by replacing structural aluminum extrusions with a custom folded-steel 'I-Beam' spine (P/N 9420-58317-00).
- **Validation:** Managed high-velocity 'War Room' coordination, utilizing a 16+ configuration test matrix to validate cooling before tooling.

DIGIDESIGN (AVID) - C|24 | Lead Mechanical Engineer | Project Curtis

2006 - 2007

| Executed a RoHS refresh of the legacy Controll24 console to eliminate a \$200/unit royalty while maintaining a \$9,995 MSRP.

- **Yield Recovery:** Salvaged pilot yield on cosmetic plastics by reducing 2.50mm warp to **<0.50mm** via 'Method A' (Vertical Hanging Fixture) and **ECO 12740**, utilizing gravity during the paint cure phase.
- **Bridge Strategy:** Executed a dual-source strategy by engaging Mass Precision for emergency manual fabrication to bypass a critical 'No-Bid' overseas production gap.
- **Margin Win:** Slashed CoGS to achieve a **51.8% Gross Margin** by eliminating legacy royalty structures and integrating 19 PCBs into a compliant architecture.

DIGIDESIGN (AVID) - D-COMMAND | Lead Mechanical Engineer | Project Danko 2004 - 2005

Engineered a mid-format control surface utilizing flagship architecture under extreme schedule compression.

- **Structural Fix:** Eliminated 5mm structural offsets and 'hammer-to-fit' assembly modes via **ECO 8000**, implementing universal bracket mounting slots to force alignment.
- **Solo Execution:** Managed 100% of mechanical documentation solo, delivering **109 unique drawings** under a 2-week deadline.
- **EMI Recovery:** Secured -2.2dB EMI headroom margin through rapid redesign of cable routing and RJ45 gasket sealing.

DIGIDESIGN (AVID) - D-CONTROL | Lead Mechanical Engineer | Project Buckley 2003 - 2004

Engineered the mechanical chassis and stand architecture for the 'ICON' flagship console.

- **Yield Rescue:** Resolved a 50% yield failure on flagship plastics via **ECO 6310**, modifying boss/hole ratios (0.37" to 0.25") to prevent strip-outs.
- **Architecture:** Directed the tooling and release of 16+ complex sheet metal assemblies under strict Class A cosmetic requirements.
- **Stabilization:** Consolidated modular stand configurations into a single unified extrusion profile, stabilizing escalating COGS.

WEBTV (MICROSOFT) - GALAXY | Principal Mechanical Architect 2001 - 2003

Architected a high-performance Home Network Gateway integrating desktop-class power (AMD K7) into a set-top form factor.

- **Thermal Architecture:** engineered a high-velocity airflow tunnel (**28.3 CFM**) to dissipate 150W, preventing Smart Card failure at 55°C.
- **Structural Stabilization:** Resolved 100% of critical chassis warpage defects by redesigning stamping features from drawn standoffs to 'Hook 29' details.

WEBTV (MICROSOFT) - CORTEZ | Product Designer 2000 - 2001

Mechanical design and complex surfacing for a reference wireless keyboard.

- **Tolerance Rectification:** Rectified a **0.05mm** accumulated tolerance error between ID pitch (19mm) and vendor tooling (19.05mm) that threatened bezel misalignment.
- **Surface Reconstruction:** Executed a surgical 43-hour reconstruction of the Pro/E database to implement organic surface lofting.
- **Optimization:** Reduced model weight to the **0.6kg target** via internal ribbing and wall thickness optimization.

WEBTV (MICROSOFT) - ELMER | Lead Mechanical Engineer 1999 - 2000

Rapid engineering of 'roadworthy' demonstration enclosures for HomeNet servers and nodes for CES 2001.

- **Thermal Risk:** Optimized convective airflow by redesigning chassis perforation patterns and implementing 'ThermaPad' gap fillers.
- **Standardization:** Enforced strict **4-40 UNC** hardware standardization, purging metric equivalents to prevent assembly line failure.
- **Delivery:** Delivered 5 functional server enclosures and 12 node units in a compressed **9-day cycle**.

MECHANISTIC | Principal / Freelance Product Designer

1999 - 2003

| *Provided expert CAD surfacing and mechanical engineering services.*

- Provided expert CAD surfacing and mechanical engineering services for clients including Apple, Motorola, and SGI.
- Designed complex organic surfaces for consumer peripherals using Pro/ENGINEER.

FROGDESIGN | Mechanical Designer

1997 - 1998

| *Bridged the gap between high-concept Industrial Design and mass production.*

- Design Fidelity: Maintained subtle surface, gap, and finish requirements while ensuring manufacturability for clients like KaVo (Dental Systems) and Newscorp (Satellite Receivers).

EARLY CAREER | Foundational Roles

1986 - 1996

| *Roles establishing the foundation of high-performance product design.*

- **Mechanistic (1993-1996):** Designed SwitchBlade inline speed skates and Sunbeam Toast Logic appliances.
- **Silicon Graphics (SGI) (1989-1993):** Performed thermal/acoustic analysis for Indigo and Indy workstations; managed prototype supply chains.
- **EP Technologies (1986-1989):** Production Supervisor for Class III medical devices (Cardiac Ablation Catheters).

EDUCATION

De Anza College

Associate's Degree, Automotive Engine Performance

AWARDS & PATENTS

- Avegant Glyph (Best of CES 2016)
- Kaleidescape Cinema One (CEPro Product of the Year)
- Motorola MP3 Player (CES Honoree) - Mechanical Design & Surfacing